

delayed market entry, and reduced revenue.

Engineering and design costs

Changes in product requirements are a common part of product development, but their impact depends largely on when they occur. During the prototyping stage, design changes are generally manageable because multiple iterations are expected as teams refine the product. However, once a product moves towards manufacturing readiness, changes become harder to implement due to fixed architecture and design decisions.

In many IoT projects, changes in requirements rarely lead to major increases in hardware costs because designers often choose chipsets that already support more features than initially required. The main challenge shifts to development effort and long-term maintainability. For this reason, requirements are typically finalised before the design is locked for production.

A key early focus is hardware-software integration. Connectivity decisions directly affect system performance, development cost, and operating expenses. As an engineer, you must select the right communication technology, such as Wi-Fi, Bluetooth Low Energy (BLE), Zigbee, LoRa, or cellular, based on application needs. Poor selection can significantly increase deployment costs, especially at scale.

Another important but often hidden engineering cost stems from the ecosystem's maturity around chipsets and connectivity solutions. While most chipsets include software development kits (SDKs), issues often appear during advanced development when documentation is incomplete, or features are not fully stable. Middleware dependencies, licensing conditions, and vendor support quality also play a major role. In some cases, newer or less established vendors may respond slowly to technical issues,

AI in product development

- AI can help automate testing and debugging by analysing signals, running test cases, and reviewing code.
- Future tools are expected to connect directly with logic analysers and development boards, helping engineers identify issues faster.
- AI can reduce development time and improve testing efficiency.
- Companies are increasingly exploring AI to understand how it can support future product development.
- AI cannot replace requirement analysis, hardware validation, vendor negotiations, or customer interactions.
- Hardware debugging often requires physical testing and engineering judgment that AI cannot provide.
- Product development still depends on human expertise, supplier support, and cross-team coordination.

causing delays as teams wait for fixes or updates. These factors can significantly extend development timelines and increase engineering effort, even when hardware costs remain unchanged.

Cloud infrastructure also begins in the engineering phase at the architectural level. Engineers define how devices communicate with the cloud, how often data is transmitted, whether real-time or batch communication is required, and what processing is handled on-device versus in the cloud. These decisions directly shape system performance, bandwidth usage, and overall architecture design.

Firmware architecture is another critical area. When firmware is tightly coupled with hardware, upgrades and migration become expensive. Modern design approaches separate application logic from hardware layers to reduce redevelopment effort when switching chipsets or platforms.

Edge analytics is also part of the engineering design stage. Processing data locally and sending only relevant outputs reduces bandwidth and cloud dependency. Techniques like Fast Fourier Transform (FFT) can help convert raw sensor data into more efficient representations.

Over-the-Air (OTA) update capability is designed during engineering for long-term flexibility. It allows remote updates, bug fixes, and feature improvements without

physical access to devices.

Regulatory compliance is also considered early in the engineering process. Selecting pre-certified modules for standards like RED (Europe) or FCC (United States) helps reduce certification effort later.

Testing and validation costs

Testing and validation are distinct and important phases of product development. Their primary purpose is to identify defects, exceptions, and performance issues before a product reaches customers. When testing is not thorough, problems may appear only after deployment, making them more expensive and time-consuming to fix.

The cost impact is particularly severe when failures involve firmware update systems or remote devices. In such cases, companies may need to send personnel to customer sites, perform manual repairs, or replace hardware. These activities add unexpected expenses that could have been avoided through comprehensive testing during development.

Testing and validation also add significant effort to IoT product development and are often a hidden source of cost and delay. Antenna tuning is one critical area, requiring specialised lab equipment and iterative measurements to ensure reliable wireless performance. Poor antenna design can lead to multiple redesign cycles.